

Time Series — Solutions

Practice Exam — Week 11: Diagnostics, Model Selection & Long Memory

EPFL, MATH-342 · Prof. S. C. Olhede

Problem 1 (5 points)

(a) Substitute the model equation into the residual: with $\hat{X}_t = \phi X_{t-1}$,

$$e_t = X_t - \hat{X}_t = (\phi X_{t-1} + \varepsilon_t) - \phi X_{t-1} = \varepsilon_t.$$

The AR structure cancels exactly, leaving the innovation. Consequently the residuals *are* the white noise, so all four checks must pass: **(i) constant zero mean** because $\mathbb{E}[\varepsilon_t] = 0$ for all t ; **(ii) constant variance** (homoscedasticity) because $\text{Var}(\varepsilon_t) = \sigma^2$ for all t ; **(iii) no autocorrelation** because $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$ (white noise); **(iv) Gaussianity** whenever the innovations ε_t are Gaussian (this is the only check that is *conditional* on a distributional assumption on ε_t rather than automatic). This is the template for all residual diagnostics: any departure from these four properties is evidence *against* the fitted model.

(b) With $n = 400$, $\sqrt{n} = 20$, so the band is

$$\left(-\frac{1.96}{\sqrt{400}}, \frac{1.96}{\sqrt{400}}\right) = \left(-\frac{1.96}{20}, \frac{1.96}{20}\right) = (-0.098, 0.098).$$

Eyeballing this ± 0.098 band lag-by-lag is a *pointwise* 95% test for each single $\hat{r}_\tau$; checking many lags simultaneously creates a multiple-testing problem (under the null, about 5% of the lags will breach the band by chance), so a single *portmanteau* statistic that aggregates many lags at once (Box–Pierce / Ljung–Box) is needed to control the overall error rate.

(c) For an AR(1) the marginal (process) variance is the geometric sum of the squared MA(∞) weights ϕ^{2j} ,

$$s_y^2 = \text{Var}(X_t) = \sigma^2 \sum_{j=0}^{\infty} \phi^{2j} = \frac{\sigma^2}{1 - \phi^2},$$

while the one-step residual variance is $s_e^2 = \text{Var}(e_t) = \text{Var}(\varepsilon_t) = \sigma^2$ by part (a). Hence

$$R^2 = 1 - \frac{s_e^2}{s_y^2} = 1 - \frac{\sigma^2}{\sigma^2/(1 - \phi^2)} = 1 - (1 - \phi^2) = \phi^2.$$

At $\phi = 0.6$: $R^2 = 0.6^2 = 0.36$. A value of 0.36 “looks low” but it is in fact the *best attainable*: a stochastic process is not perfectly predictable, so even the true, optimal model leaves the fraction $1 - \phi^2$ of the variance unexplained — a low R^2 is therefore no indictment. Conversely, fitting a richer AR(4) to the same data can only *increase* the in-sample R^2 (adding parameters never reduces explained variance), so a higher R^2 is no endorsement of the larger model. The correct tool for choosing between AR(1) and AR(4) is an information criterion (AICc/BIC) or a Ljung–Box test on the AR(1) residuals, *not* R^2 .

Problem 2 (5 points)

(a) Subtract the two criteria and put the bracket over a common denominator:

$$\begin{aligned} \text{AICc} - \text{AIC} &= 2k \left(\frac{n}{n - k - 1} - 1 \right) = 2k \cdot \frac{n - (n - k - 1)}{n - k - 1} \\ &= 2k \cdot \frac{k + 1}{n - k - 1} = \frac{2k(k + 1)}{n - k - 1} (> 0). \end{aligned}$$

The correction is always positive (AICc penalises complexity more heavily). The two criteria are approximately equivalent exactly when this term is small:

- if k is **fixed** and $n \rightarrow \infty$, then $\frac{2k(k+1)}{n-k-1} \rightarrow 0$, so $\text{AICc} \rightarrow \text{AIC}$;
- if $k = k_n$ grows with n , the gap is still negligible provided n is large relative to k_n^2 , e.g. $k_n^2/n \rightarrow 0$. If k grows too fast ($k_n^2/n \not\rightarrow 0$) the gap need not vanish and AICc stays meaningfully larger.

In words: *large samples relative to the number of parameters* make the two agree; small samples or richly parameterised models keep AICc strictly more conservative.

(b) If m of the $p + q$ AR/MA coefficients are fixed (not estimated), only $p + q - m$ of them are estimated. Adding the innovation variance σ^2 (and, for a mean-zero model, no separate mean parameter), the estimated-parameter count is

$$\boxed{k = p + q + 1 - m}.$$

Substituting into the AICc formula,

$$\text{AICc}(\theta) = -2 \ln L(\theta | y) + 2(p + q + 1 - m) \frac{n}{n - (p + q + 1 - m) - 1}.$$

Fixing coefficients to zero lowers k , which both shrinks the penalty and (in a subsequent Ljung–Box test) raises the residual degrees of freedom.

(c) With $n = 30$:

- Full AR(4): $k = p + q + 1 = 4 + 0 + 1 = 5$, so

$$2k \frac{n}{n - k - 1} = 2 \cdot 5 \cdot \frac{30}{30 - 5 - 1} = 10 \cdot \frac{30}{24} = 12.5.$$

- Constrained AR(4), $m = 2$ fixed: $k = 4 + 0 + 1 - 2 = 3$, so

$$2k \frac{n}{n - k - 1} = 2 \cdot 3 \cdot \frac{30}{30 - 3 - 1} = 6 \cdot \frac{30}{26} = \frac{180}{26} \approx 6.923.$$

The constrained model carries the smaller penalty by $12.5 - 6.923 = 5.577$. Writing $\text{AICc} = -2 \ln L + \text{penalty}$, AICc prefers the full model only when

$$[-2 \ln L_{\text{full}} + 12.5] < [-2 \ln L_{\text{constr}} + 6.923],$$

i.e. when $-2 \ln L_{\text{constr}} - (-2 \ln L_{\text{full}}) > 12.5 - 6.923 = 5.577$. So the constrained model's $-2 \ln L$ may exceed the full model's by up to $\boxed{\approx 5.58}$ before AICc switches its preference to the full AR(4); equivalently, the two extra estimated coefficients must improve the log-likelihood by more than ≈ 2.79 to be worth keeping under AICc .

Problem 3 (5 points)

(a) With $n = 200$ we have $n(n+2) = 200 \cdot 202 = 40400$, and $m = 5$. The Ljung–Box statistic is

$$Q_5 = 40400 \left(\frac{0.05^2}{199} + \frac{(-0.08)^2}{198} + \frac{0.12^2}{197} + \frac{0.04^2}{196} + \frac{(-0.10)^2}{195} \right).$$

Term by term,

$$\begin{aligned} \frac{0.0025}{199} &= 1.256 \times 10^{-5}, & \frac{0.0064}{198} &= 3.232 \times 10^{-5}, & \frac{0.0144}{197} &= 7.310 \times 10^{-5}, \\ \frac{0.0016}{196} &= 8.163 \times 10^{-6}, & \frac{0.0100}{195} &= 5.128 \times 10^{-5}. \end{aligned}$$

Summing the five terms gives

$$1.256 \times 10^{-5} + 3.232 \times 10^{-5} + 7.310 \times 10^{-5} + 8.163 \times 10^{-6} + 5.128 \times 10^{-5} = 1.7743 \times 10^{-4}.$$

Hence

$$Q_5 = 40400 \times 1.7743 \times 10^{-4} \approx 7.17.$$

(b) For the ARMA(2, 1) fit, $p = 2$ and $q = 1$, so the degrees of freedom are

$$\text{df} = m - p - q = 5 - 2 - 1 = 2.$$

The 5% critical value is $\chi_{2,0.95}^2 = 5.991$. Since $Q_5 \approx 7.17 > 5.991$, we **reject** H_0 at the 5% level: there is significant residual autocorrelation up to lag 5, so the ARMA(2, 1) model is *inadequate* and should be revised.

(c) For an AR(1) fit, $p = 1$, $q = 0$, so

$$\text{df} = m - p - q = 5 - 1 - 0 = 4,$$

and the critical value rises to $\chi_{4,0.95}^2 = 9.488$. Now $Q_5 \approx 7.17 < 9.488$, so we would **fail to reject** H_0 (the test deems the residuals adequately white). The *same* numbers thus give opposite conclusions purely through the degrees of freedom: subtracting the correct number of estimated ARMA parameters ($p + q$) is essential, because using $\text{df} = m$ would inflate the critical value (here $\chi_{5,0.95}^2 = 11.07$) and make the test reject far too rarely, masking genuine misspecification.

Problem 4 (5 points)

(a) Iterating the causal AR(1) recursion gives the MA(∞) representation $X_t = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}$, so the weights are

$$\psi_j = \phi^j, \quad j = 0, 1, 2, \dots$$

For the best linear (here, since Gaussian, also overall best) predictor, write $X_{n+h} = \phi^h X_n + \sum_{j=1}^h \phi^{h-j} \varepsilon_{n+j}$ and take the conditional expectation given $X_1, \dots, X_n$. Every future innovation $\varepsilon_{n+1}, \dots, \varepsilon_{n+h}$ has conditional expectation 0 (it is uncorrelated with the data, and zero-mean), leaving

$$X_{n+h}^n = \mathbb{E}[X_{n+h} \mid X_1, \dots, X_n] = \phi^h X_n.$$

(b) The h -step forecast error is $e_n(h) = X_{n+h} - X_{n+h}^n = \sum_{j=0}^{h-1} \psi_j \varepsilon_{n+h-j} = \sum_{j=0}^{h-1} \phi^j \varepsilon_{n+h-j}$, a sum of uncorrelated terms, so

$$\text{Var}(e_n(h)) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2 = \sigma^2 \sum_{j=0}^{h-1} \phi^{2j} = \sigma^2 \frac{1 - \phi^{2h}}{1 - \phi^2}.$$

As $h \rightarrow \infty$, since $|\phi| < 1$ we have $\phi^h \rightarrow 0$, so the point forecast $X_{n+h}^n = \phi^h X_n \rightarrow 0$, the process mean; and $\phi^{2h} \rightarrow 0$, so

$$\text{Var}(e_n(h)) \rightarrow \frac{\sigma^2}{1 - \phi^2} = \gamma_0,$$

the marginal variance of the AR(1). Far-ahead forecasting thus reverts to “the unconditional distribution”: point forecast $\rightarrow$ mean, error variance $\rightarrow \gamma_0$.

(c) With $\phi = 0.6$, $\sigma^2 = 4$, $X_n = 5$:

$$X_{n+2}^n = \phi^2 X_n = 0.36 \times 5 = 1.8.$$

The two-step forecast-error variance is

$$\text{Var}(e_n(2)) = \sigma^2(1 + \phi^2) = 4(1 + 0.36) = 4 \times 1.36 = 5.44,$$

so the standard error is $\sqrt{5.44} \approx 2.332$. A 95% Gaussian prediction interval for X_{n+2} is

$$X_{n+2}^n \pm z_{0.975} \sqrt{\text{Var}(e_n(2))} = 1.8 \pm 1.96 \times 2.332 = 1.8 \pm 4.571,$$

i.e.

$$\boxed{X_{n+2} \in (-2.77, 6.37) \text{ with 95\% confidence.}}$$

(As a check, $\gamma_0 = \sigma^2/(1 - \phi^2) = 4/0.64 = 6.25$, and indeed $\text{Var}(e_n(2)) = 5.44 < 6.25$, with the gap closing as h grows.)